Khot Om Dhanaji

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.52
Shraddha Junior College of Science and Commerce Ichalkaranji	Class XII (Maharashtra State Board)	2023	90.00%
Shri. Yashawantarao Ghatage Highschool Kagal	Class X (Maharashtra State Board)	2021	99.60%

WORK EXPERIENCE

Software Development Engineer Intern | Kotak Mahindra Bank

Project 1: Bill-Fetch Service

- Developed core components of the **Bill-Fetch service**, processing **10M+ users** daily through automated due-bill retrieval and tracking within Kotak's digital banking ecosystem.
- Implemented the **DTO, DAO and database schema layers**, including the Executor schema and DAO, to support bill-fetch persistence and execution tracking.
- Developed scheduled **Cron-based Producer workflows** to generate Bill-Fetch instances efficiently, tracking task execution, completion and failure states using the Executor design.
- Validated bill-fetch workflows through unit, integration and end-to-end testing, including phased production testing; analyzed execution outcomes, logs and production data using AWS Athena to investigate failures.

Project 2: Notification Service

- Developed the Notification Service's **Schema** and **DAO** layers and configured reusable notification message templates for transaction, payment, autopay and bill-due alerts based on product requirements.
- Implemented **event-driven notification processing** using AWS EventBridge and SQS, supporting the publishing and handling of **millions of daily notifications**, while integrating payment-status updates with the **Refund Service** to streamline refund processing.

SKILLS

- **Programming Languages and Fundamentals:** C++, Java, Python, SQL, OOP, OS, Data Structures and Algorithms
- **Backend & System Design:** Spring Boot, REST APIs, Microservices, Redis, Rate Limiting, Caching, Reverse Proxy, Distributed Systems, Producer-Consumer Architecture, Executor Design Pattern
- **Cloud, DevOps & Tools:** AWS (S3, Lambda, DynamoDB, EventBridge, SQS, Athena), Docker, Git/GitHub, Postman
- **Observability & Testing:** Kibana, Grafana, JUnit 5, Mockito, Unit & Integration Testing

PROJECTS

- **Smart API Gateway -- Enterprise API Gateway** GitHub Link

Java 21, Spring Boot, Spring Cloud Gateway, Redis, Resilience4j, Docker, Prometheus, Grafana

- Developed a centralized **API Gateway** using Spring Cloud Gateway to handle dynamic routing, reverse proxying, JWT authentication and runtime route configuration across **4** microservices.
- Engineered **Redis-backed Token Bucket rate limiting** and TTL-based response caching, cutting average response time from **80 ms to 10 ms** for cached endpoints in load tests.
- Implemented **round robin** load balancing across replicated services, added **circuit breakers and exponential-backoff retries** (Resilience4j) with Prometheus/Grafana monitoring; sustained **7,000**+ **req/sec** at **P99 latency under 20 ms** under simulated load (using k6).

- **LiteQueue -- Redis-Backed Asynchronous Job Queue** GitHub Link

Java 21, Spring Boot, Redis, Docker, Maven, JUnit 5, Mockito

- Designed a **Redis-backed job queue** using Redis Lists for FIFO processing and Sorted Sets for delayed/scheduled jobs, with a fixed worker pool for concurrent execution.
- Engineered **startup-time crash recovery** to detect and **re-queue jobs** left in-flight after a worker failure, recovering around **98%** jobs correctly in controlled failure testing.
- Built **Strategy/Registry patterns** for pluggable job executors and a Dead Letter Queue for failed jobs; validated with **85% unit test coverage** (JUnit 5, Mockito)

- **AutoCalendar AI -- Autonomous Email-to-Calendar Assistant** GitHub Link

Python, MongoDB, Gmail API, Google Calendar API, OAuth 2.0, LLM APIs

- Built an autonomous email-to-calendar assistant for **working professionals and students**, automating event and deadline extraction from Gmail and synchronization with Google Calendar to reduce manual reminder creation.
- Developed an **asynchronous background-worker pipeline** integrating LLM APIs to parse event title, date, time and venue, correctly extracting event details through prompt engineering and structured JSON parsing across test scenarios.
- Implemented **OAuth 2.0-based multi-user sync** with MongoDB, using **idempotent tracking** to prevent duplicate calendar events, validated across **3 test accounts** over **5+ polling cycles** with zero duplicate injections.

ADDITIONAL INTERESTS

- **Sanskrit Language:** Explored Sanskrit language fundamentals through the creative arts course Samskritam: Sambhashana Praveshah.